

Monday: Ch 5: Question 13, and Problem 44

Tuesday: Ch 6: Question 1 and Problem 47 a,b

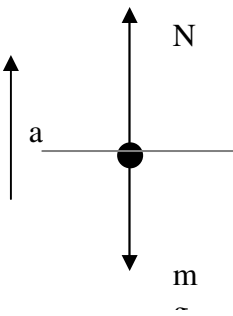
Wednesday: Ch 6: Question 2 and Problem 47 a,c

Chapter 5 Question 13:

- a) Normal force increases.
b) Normal force decreases.

Chapter 5 Problem 44

44. (a) The term “deceleration” means the acceleration vector is in the direction opposite to the velocity vector (which the problem tells us is downward). Thus (with +y upward) the acceleration is $a = +2.4 \text{ m/s}^2$.



Newton's second law states:

$$\begin{aligned}\sum F_y &= ma_y = T - mg \\ T &= ma_y + mg \\ m &= \frac{T}{(a_y + g)} = \frac{93 \text{ N}}{2.4 \text{ ms}^{-1} + 9.8 \text{ ms}^{-1}} \\ m &= 7.6 \text{ kg}\end{aligned}$$

which yields $m = 7.6 \text{ kg}$ for the mass.

(b) Repeating the above computation (now to solve for the tension) with $a = +2.4 \text{ m/s}^2$ will, of course, lead us right back to $T = 93 \text{ N}$. Since the direction of the velocity did not enter our computation, this is to be expected.

Chapter 6:**Question 1**

a) decrease, b) decrease, c) increase, d) increase, e) increase

Question 2

a) decrease, b) decrease, c) decrease, d) decrease, e) decrease

Chapter 6 Problem 47

(a) The period of rotation is the time taken for 1 rotation:

$$T = \frac{1 \text{ rotation}}{\text{velocity}} = \frac{2\pi r}{v} = \frac{2\pi(12 \text{ m})}{5.5 \text{ ms}^{-1}} = 11 \text{ s}$$

(b) The situation is similar to that of Sample Problem – “Vertical circular loop, Diavolo,” but with the normal force direction reversed. At the highest point in the rotation, the acceleration (due to circular motion) is directed towards the centre of the circular path – i.e. vertically downward. The free body diagram then looks like this:



$$\sum F_y = -ma_y = N - mg \quad (a_y \text{ is negative therefore the negative sign})$$

$$N = mg - m\frac{v^2}{r} = m\left(g - \frac{v^2}{r}\right)$$

$$N = 582 \text{ N}$$

(c)



Similarly we now have the net acceleration still pointed towards the centre of the circle, but now upwards. The other two forces are the same as before. Now from Newton's 2nd law:

$$\sum F_y = ma_y = N - mg$$

$$N = m\frac{v^2}{r} + mg = m\left(\frac{v^2}{r} + g\right)$$

$$N = 986 \text{ N}$$